

IP Address/Logical Address

IPv4 = 32 bits(4 bytes)

IPv6 = 128 bits(16 bytes)

2^0

2^1

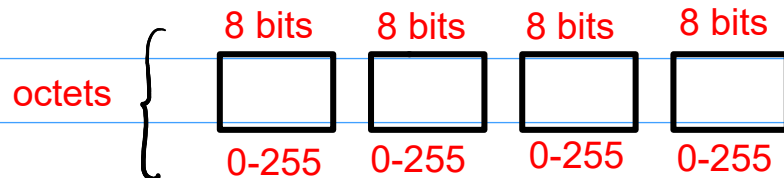
..

..

..

..

2^{31}



$2^8 = 256(0-255)$

Dotted
Decimal
Notation

Minimum IP : 0.0.0.0

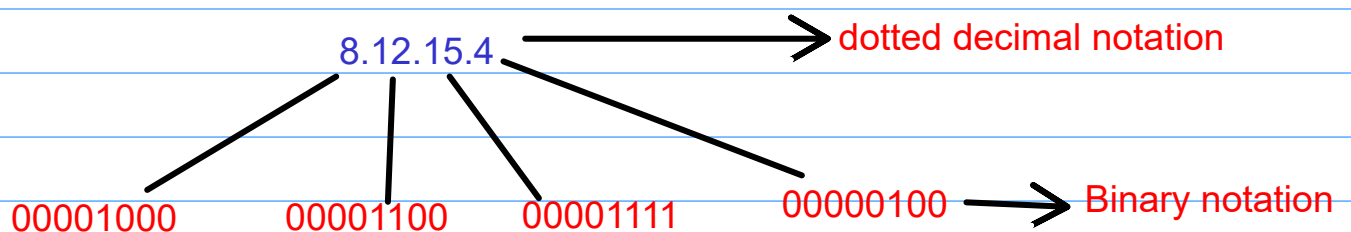
Maximum IP : 255.255.255.255

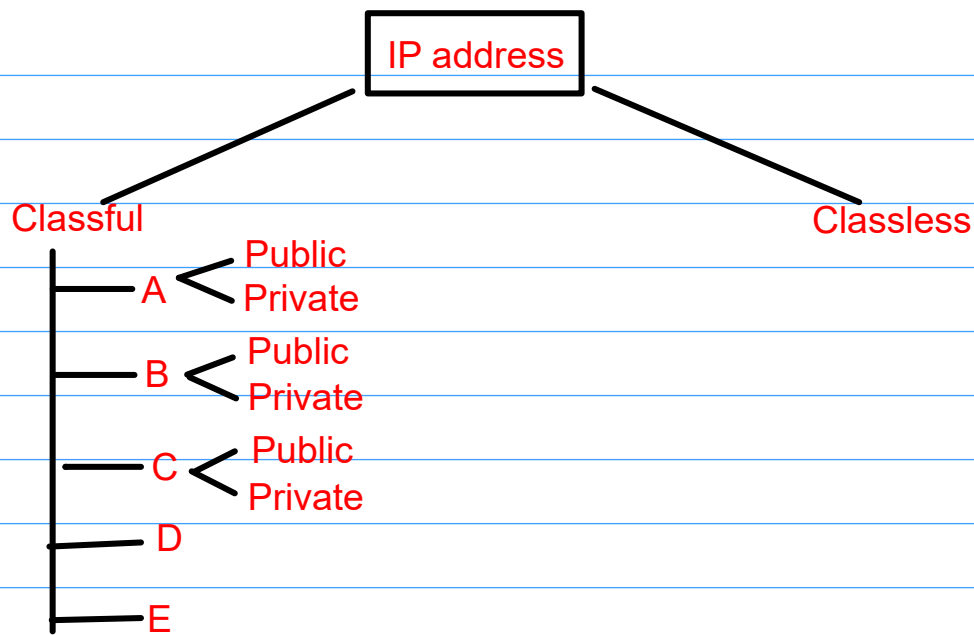
Binary

Notation

Minimum IP : 00000000 00000000 00000000 00000000

Maximum IP : 11111111 11111111 11111111 11111111





First octet(first byte) of any IP Address represents the CLASS		8 bits	8 bits	8 bits	8 bits
		0-255	0-255	0-255	0-255
Class A	0 - 127	0.0.0.0 to 127.255.255.255			
Class B	128 - 191	128.0.0.0 to 191.255.255.255			
Class C	192 - 223	192.0.0.0 to 223.255.255.255			
Class D	224 - 239	224.0.0.0 to 239.255.255.255			
Class E	240 - 255	240.0.0.0 to 255.255.255.255			

eg: 120.13.45.67 valid Class A

193.23.47.89 valid Class C

246.12.24.65 valid Class E

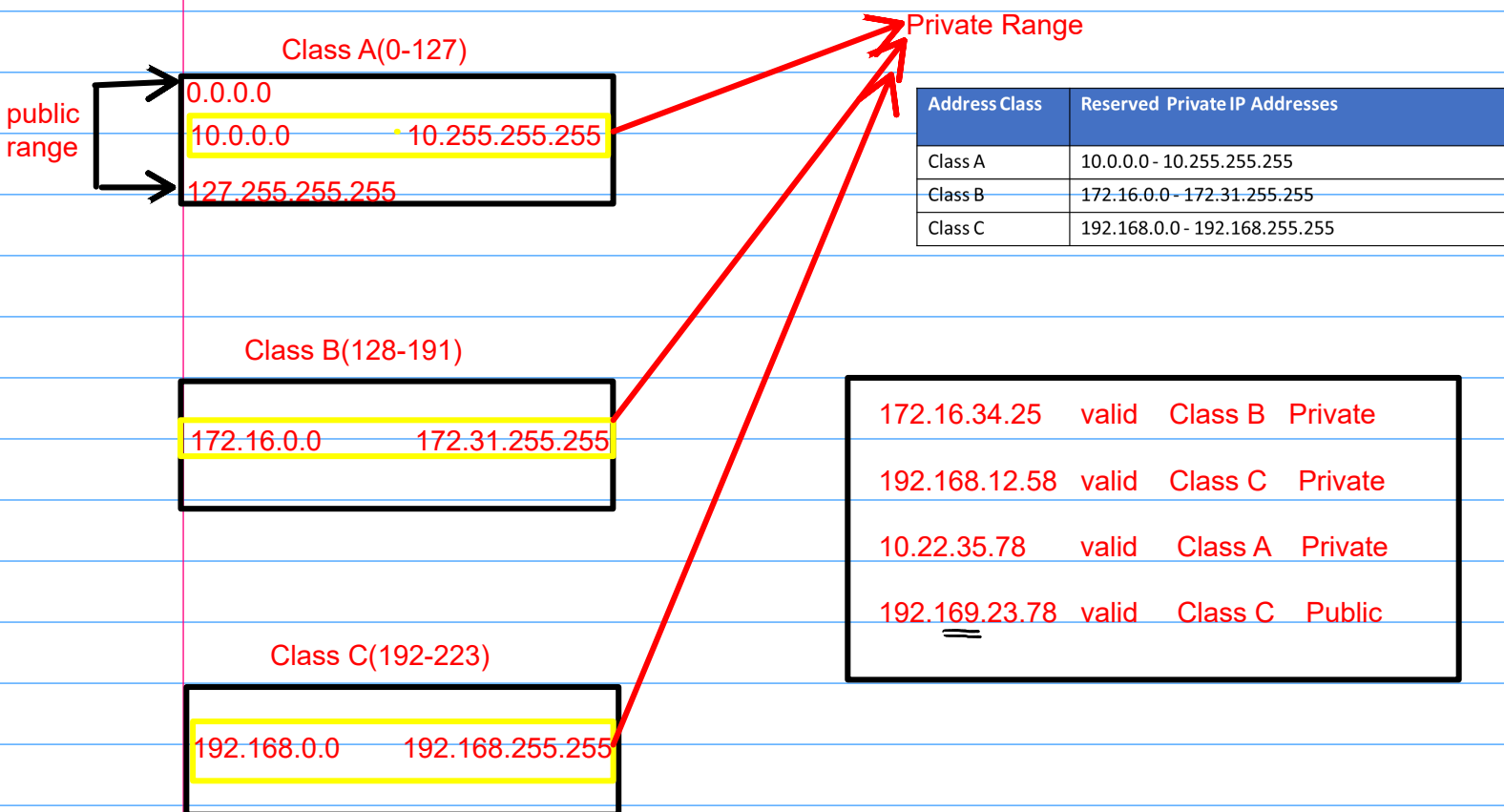
129.12.67.266 invalid

0 - 255

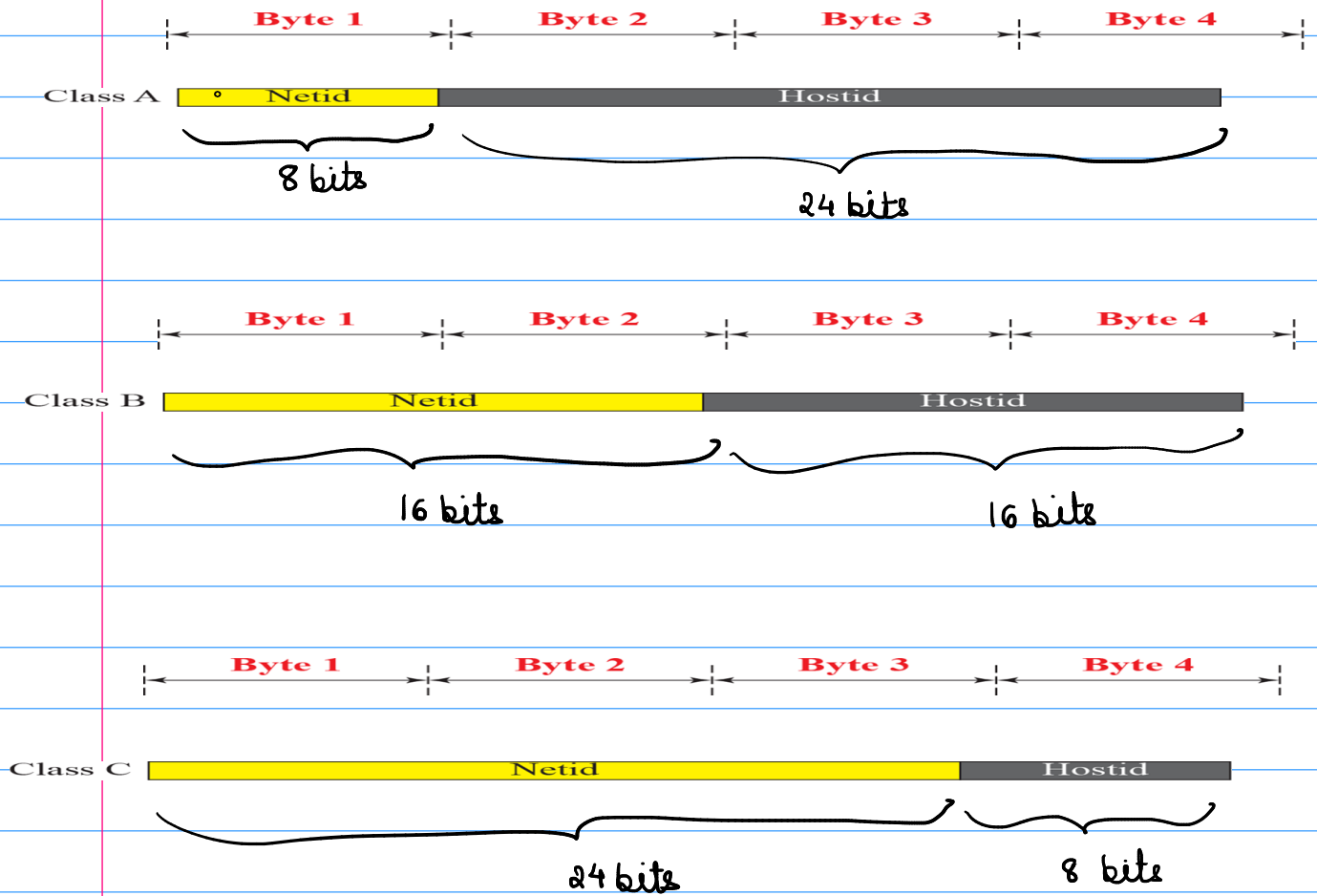
0	10	110	1110	1111
Class A(0-127)	Class B(128-191)	Class C(192-223)	Class D(224-239)	Class E(240-255)
0 0000 0000	128 1000 0000	192 1100 0000	224 1110 0000	240 1111 0000
1 0000 0001	129 1000 0001	193 1100 0001	225 1110 0001	241 1111 0001
2 0000 0010	...	...	..	..
...	...	...	..	..
...	...	...	..	..
...	...	...	..	..
...	...	...	..	..
...	...	...	..	..
127 0111 1111	191 1011 1111	223 1101 1111	239 1110 1111	255 1111 1111

NOTE :

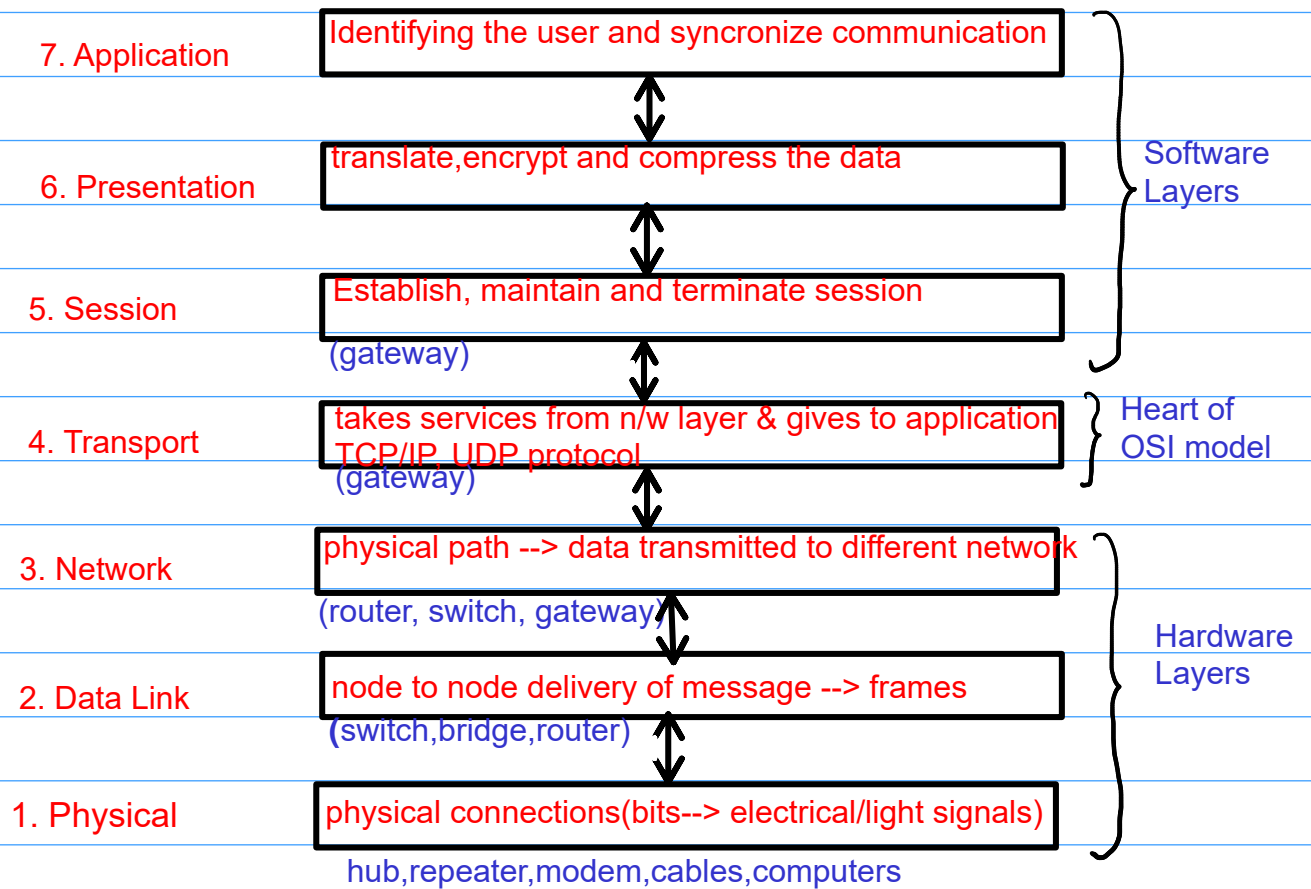
- 127.x.x.x such IP address which starts with 127 is called as loop back series
- used for self testing
- 127.0.0.1 --> ping yourself
- 127.0.0.1 --> universal IP address which is default configured.
You cannot configure it.
- PING --> Packet Internet Groper (its a tool used to troubleshoot network issues)
- IANA --> Internet Assigned Numbers Authority



Netid and Hostid of A,B and C Classes



OSI Model(7 Layers)



TCP(Transmission Control Protocol)	UDP(User Datagram Protocol)
1. connection oriented	1. connectionless
2. Less speed(ACK)	2. Faster speed
3. Data quality is of importance	3. Data quality dos not matter
4. eg: email, file transfer	4. eg: online gaming, OTT